11 THE GLOBAL POSITIONING SYSTEM (GPS)

THE SPACE SEGMENT

1. The TRANSIT Doppler satellite positioning system was released for commercial use in 1967 and has since been widely used. GPS technology represents an improvement over the earlier TRANSIT system in many respects. Which of the following statements correctly describes some of those improvements?

(A) The orbits of the satellites of the TRANSIT system are much more susceptible to variations in local gravity fields and atmospheric drag than are the satellites of the GPS.

(B) The use of the TRANSIT system requires that line-of-sight be maintained between the stations of a network, while GPS does not.

(C) The timing technology in GPS is much improved from that which was available when the TRANSIT system was established.

(D) Both A and C are true.

Solution: Neither the TRANSIT system nor GPS requires a line-of-sight between stations for successful satellite positioning. However, GPS has brought many improvements over the TRANSIT system. For example, the signals from the TRANSIT satellites are much more likely to be affected by ionospheric delay than are those of GPS. The submeter accuracy of the TRANSIT system has been outstripped by GPS, and observations using GPS technology are much more efficient than those with the TRANSIT system.

Answer is (D)

2. What is the name given to the GPS satellites?

(A) OSCAR
(B) NOVA
(C) NAVSTAR
(D) ATLAS

Solution: The GPS satellites are named NAVSTAR, an acronym for Navigatior System by Timing And Ranging. The NAVSTAR number, such as NAVSTAR 8 indicates the launch sequence of the space vehicle to which it is assigned.

Answer is (C)

3. Which of the following orbital parameters best describes the final plannec constellation of GPS satellites?

(A) six orbital planes inclined 55° with respect to the celestial equator
(B) nearly circular orbits at 26,000 km
(C) nearly circular orbits at 1100 km
(D) both A and B

Solution: The full constellation of GPS satellites is designed to provide fou satellites above the horizon everywhere on the earth's surface, 24 hours a day The 26,000-km orbit provides an orbital period of nearly 12 hr, though eacl satellite will rise 4 min earlier each day. If earlier propositions of three orbita planes of six satellites each had been adopted, more areas of reduced coverag would have resulted. The current constellation plan does have outages—times o inferior positioning accuracy—but these periods are generally less than 10 min.

Answer is (D)

4. What is the mask angle in GPS, and what is its significance?

(A) Reflected signals received by a GPS antenna are eliminated by a ground plane that defines the mask angle.

(B) The mask angle is used in special circumstances to eliminate the reception of signals from satellites with unstable clocks.

(C) The mask angle is the zenith at which a particular antenna type ceases to receive signals efficiently and is different for each type.

(D) The mask angle is a vertical angle of 10°–20° from the observer's horizon, below which signals from the GPS satellites are considered less reliable.

Solution: The signal from NAVSTAR satellites must pass through a thicker layer of the atmosphere when the satellites are close to the observer's horizon. Variations of the signal, called *tropospheric errors*, are especially unpredictable within the mask angle area, the signals are generally not considered reliable in that region.

Answer is (D)

5. What are the positioning signal frequencies used in GPS?

(A) 150 MHz and 400 MHz

(B) 1832.12 MHz and 3236.94 MHz

(C) 420.9 MHz and 585.53 MHz

(D) 1575.42 MHz and 1227.60 MHz

Solution: The two frequencies broadcast by GPS satellites for positioning are 1575.42 MHz, known as L1, and 1227.60 MHz, known as L2. Both are derived from the fundamental frequency 10.23 MHz. L1 is created by the multiplier 154 and L2 by the multiplier 120.

Answer is (D)

6. The carriers L1 and L2 are modulated by two PRNs, or pseudo-random noise codes. What are the names given to these codes?

(A) f_0 code and f_1 code
(B) Ψ code and Δ code
(C) P code and C/A code
(D) IDOT code and IODE code

Solution: The PRN codes are known as the *C/A code* and the *P code*. The P code is the more precise code, and is the principal code for navigation. Both codes consist of positive and negative ones (+1 and −1) created by modulating the carrier with phase shifts. A −1 is indicated by a 180° shift in phase and a +1 by no phase shift at all. Both carriers, L1 and L2, are modulated by the P code; the C/A code is found only on L1. The C/A code, or coarse/acquisition code, is emitted at a frequency of 1.023 MHz; the P code has a frequency of 10.23 MHz.

Answer is (C)

7. The PRN codes repeat themselves over specific intervals of time. What are the periods of those cycles, and what is their significance?

(A) The P code has a 38-week period before it repeats itself, but each week of that cycle is assigned to a different satellite. The C/A code repeats itself every millisecond and each satellite transmits a different C/A code, allowing a receiver to distinguish signals received simultaneously from different satellites.
(B) The P code repeats itself every second, during which time it has over 10 million opportunities to switch from a +1 to a −1. The C/A code repeats itself every second and is, therefore, quite easily acquired.
(C) The P code repeats itself every nanosecond. The C/A code repeats itself every week and can be used as a moderately accurate navigation signal as well as a method of rapidly acquiring the P code.
(D) All of the above are true at selected intervals determined by the GPS control segment.

Solution: The C/A code is designed for rapid acquisition by a receiver. Since it is synchronized with the P code, it provides a good way to acquire the more precise code as well. The repetition cycle of the C/A code is very short: 1 ms. Many surveying applications of GPS use only the C/A code. The P code is used to assign a PRN number to each GPS satellite, a number that differs from its NAVSTAR or launch sequence number. This PRN number indicates the weekly code sequence assigned to a particular satellite. For example, PRN 12 would consistently be assigned to the 12th week of the 38-week-long P code sequence. This system allows for the transmission of mutually exclusive P code sequences from each space vehicle.

Answer is (A)

8. A navigation message is modulated onto both of the L1 and L2 carrier frequencies. What is the frequency of the navigation message, and what is its significance?

(A) The frequency of the navigation message is 100 MHz, and the message contains information on the position of the GPS receiver.

(B) It is broadcast at 50 Hz and contains information on the status of both the particular satellite and the system as a whole.

(C) The navigation message is broadcast at 1000 MHz, and contains information on the clock corrections needed for the satellite timing systems and the ephemerides for all of the GPS satellites.

(D) The frequency of the navigation message is 450 Hz, and the message contains information on the health of all of the GPS space vehicles as well as messages, flags, and ionospheric delay parameters.

Solution: The navigation code, or satellite message, is transmitted at a frequency of 50 Hz, and has information on the system status, clock behavior, GPS time, the ephemerides of the satellites, etc. The full message is 1500 bits long and is divided into five subframes of 300 bits each.

Answer is (B)

9. The Block II GPS satellites are rather large vehicles, 845 kg in final orbit. The satellites are equipped to perform several general functions. Which of the following statements correctly characterizes some of those functions?

(A) The satellites receive transmissions from the receivers of GPS users and transmit timing and positioning information to the receivers of GPS users.

(B) The satellites store information received from the GPS ground control stations and do some basic data processing on board. The satellites also maintain very accurate time.

(C) The satellites are equipped with thrusters to maneuver in orbit and have a life expectancy of 7.5 years.

(D) Both B and C are true.

Solution: GPS is a passive system, meaning that the satellites do not receive any transmissions from the receivers of GPS users. The only information received by the satellites is transmitted by the official government GPS control ground stations, or upload stations.

Answer is (D)

10. Timekeeping is critical to the operation of the GPS system. Which of the following statements best describes the devices used to maintain accurate time for positioning with GPS?

(A) Each GPS Block II satellite is equipped with two high-precision oscillators, and GPS receivers do not have on-board clocks.

(B) Most GPS receivers have quartz crystal oscillators as their internal clock, and each Block II GPS satellite has four atomic oscillators.

(C) Most GPS receivers employ hydrogen masers as their timekeeping device, and each Block II GPS satellite uses quartz crystal oscillators as its frequency standard.

(D) Each GPS Block II satellite and most GPS receivers employ hydrogen masers as their timekeeping devices.

Solution: Every Block II satellite has four atomic high-precision oscillators, two cesium and two rubidium, the former being the more accurate and the latter the less expensive. Only one of these four clocks is used at a time; the others provide redundancy. Most GPS receivers employ oven-controlled quartz crystal oscillators as their time standard. The oscillators depend on the piezoelectric effect for their capabilities. They are inexpensive, compact, and consume very little power.

Answer is (B)

CONTROL SEGMENT

11. The GPS control segment consists of several government facilities that are responsible for operating the global positioning system. Which of the following procedures would not be executed by the control segment?

(A) continuous tracking of all GPS satellites
(B) creating updated navigation messages
(C) providing precise ephemeris data to the user community
(D) uploading corrected ephemeris data to the satellites

Solution: The control segment monitors the satellites and produces corrected navigation messages that it uploads on a daily basis. However, it is not the responsibility of the control segment to provide post-computed precise ephemeris data to the civilian-user community. Several organizations, both public and private, have announced plans to provide this service.

Answer is (C)

12. What is the meaning of selective availability in GPS?

(A) the deliberate degradation of the GPS signal
(B) the encryption of the P code transmissions
(C) the licensing of all GPS receivers by the federal government
(D) the shutdown of selected satellites by the control segment

Solution: The Departments of Defense and Transportation developed the Federal Radionavigation Plan. This plan includes the ability to limit the availability of point positioning services superior to the 100-m level to military and specific nonmilitary use. Selective availability (SA) is one way of degrading GPS accuracy by introducing clock errors into the navigation message transmission, providing the user with a false pseudo-range. However, SA has no adverse effect on positioning accomplished by differential GPS techniques.

Answer is (A)

13. The Joint Program Office has specified a three-dimensional point positioning limit that should normally be available to P code users. Consider also the degradation of two-dimensional civilian accuracy with selective availability. What is the maximum degradation allowed in these GPS point positioning accuracies at 2σ confidence levels?

(A) 16 m with full P code access
100 m with C/A code

(B) 100 m with full P code access
200 m with C/A code

(C) 1 m with full P code access
10 m with C/A code

(D) 0.01 m with full P code access
1 m with C/A code

Solution: It must be emphasized that these general accuracies are for point positioning only and do not apply to differential or relative positioning methods in GPS. It is impossible for the control segment to perfectly predict the position of the satellite in the navigation message it uploads. Therefore, the navigation message ephemeris information read by the receiver from the GPS satellites inevitably introduces errors into the position determination. However, maximum limits of this degradation have been set. Generally, the degradation of available point positions should not be more than 16 m for a user with full P code access, and 100 m for C/A code users at a 2σ confidence level.

Answer is (A)

14. The remote monitoring stations transfer their data to the Master Control Station, where computed satellite updates are generated and sent to the upload stations. Where is the Master Control Station located?

(A) Hawaii
(B) Ascension Island
(C) Diego Garcia
(D) Colorado Springs, Colorado

Solution: The Master Control Station is in Colorado Springs, Colorado. Monitoring stations are located at all of the other places listed, and there are upload facilities at all but Hawaii.

Answer is (D)

GPS RECEIVERS

15. What is the purpose of a ground plane on a GPS antenna?

(A) The ground plane of a GPS antenna provides a convenient reference for its correct polarization.
(B) The ground plane of a GPS antenna improves its gain and coverage.
(C) The ground plane of a quadrifilar or volute antenna assists the adjustment of its element phasing.
(D) The ground plane of an antenna reduces the interference from reflected signals.

Solution: Some GPS antenna types are more susceptible than others to interference from GPS signals that bounce off nearby objects or the ground. The reflected signals, called *multipath*, present a confusing contradiction to the signals received directly from the satellite. The ground plane of a GPS antenna reduces the effect of multipath by blocking most of these reflected signals.

Answer is (D)

16. GPS receivers are sometimes distinguished by the number of channels they include. What sorts of channels are involved, and what is their function?

(A) The channels of a GPS receiver are instructions in the software that define the phase center of the antenna.

(B) The GPS receiver must have the capability of isolating signals received simultaneously from several satellites. This isolation is accomplished through the use of dedicated channels.

(C) The channels of a GPS receiver are the number of its intermediate frequency (IF) stages.

(D) None of the above are true.

Solution: Channels are dedicated to particular satellites at particular times within a GPS receiver. Through the use of software, hardware, or a combination of both, the separate satellites' signals can be distinguished by their unique codes or their particular Doppler shifts.

Answer is (B)

17. What is an advantage of a dual-frequency receiver over a single-frequency receiver?

(A) Much of the error introduced by ionospheric delay can be eliminated with a dual-frequency receiver.

(B) Dual-frequency receivers are cheaper than single-frequency receivers.

(C) Dual-frequency receivers use the C/A code and single-frequency receivers must use the P code.

(D) Over long baselines, there is no significant advantage of a dual-frequency receiver over a single-frequency receiver.

Solution: The time delay of a signal passing through the ionosphere is proportional to the inverse of the square of the frequency of that signal. The higher the frequency, the shorter the time delay. In other words, L1 passes through the

ionosphere faster than L2, allowing fairly accurate calculation and near elimination of the ionospheric delay using dual-frequency GPS receivers. This is perhaps the main advantage of a dual-frequency GPS receiver over a single-frequency receiver.

Answer is (A)

18. GPS receivers are generally divided into those that use channel squaring, sometimes known as *codeless receivers*, and those that use code correlation. Which of the following statements correctly identifies some of the advantages and disadvantages of the two categories?

(A) Code correlation receivers cannot correct for ionospheric delay without access to the P code and are therefore inferior to codeless receivers over long baselines.

(B) Codeless receivers have no access to the broadcast navigation message. Their clocks must be synchronized externally and the satellite orbital data must also be provided to them externally. Therefore, a purely codeless receiver is somewhat more cumbersome to use than a code correlation receiver.

(C) Codeless receivers have a much stronger signal-to-noise ratio than code correlation receivers and are less expensive.

(D) Both A and B are true.

Solution: The squaring of the carrier frequency received from the GPS satellite provides a method of using both L1 and L2 simultaneously to improve positioning accuracy over long baselines. However, the process strips the signal of its codes and navigation messages since all phase inversions are immediately eliminated. Code correlation receivers retain the codes and messages, but most civilian receivers of this type work only with L1, since the C/A code is not available on L2. However, code correlation receivers enjoy a certain advantage over the codeless receivers in signal-to-noise ratio since squaring the carrier frequency also squares the noise. The single-frequency code correlation receiver is the less expensive of the two.

Answer is (D)

19. Which of the following would be available from a six-channel GPS receiver?

(A) the ability to track six satellites using sequencing 20 ms long, which is the length of one bit of the satellite message

(B) the ability to track six satellites by switching from one to another with a microprocessor

(C) the ability to track six satellites continuously with hardware dedicated to one signal each

(D) the ability to sequence through the signals from six satellites without spanning any bit boundary

Solution: Hardware that is dedicated to one satellite signal and can track that signal continuously constitutes a channel. Some GPS receivers track several satellites by reading each signal individually in a sequence. These receivers may track four, six, or more satellites sequentially, but not any one continuously.

Answer is (C)

20. Which of the following is characteristic of a multiplexing receiver in GPS?

(A) It is a hybrid receiver combining both code-correlation tracking and codeless tracking.

(B) It has the receiver computer and the GPS antenna combined into one unit.

(C) It is capable of tracking only with the P code.

(D) It tracks by sequencing, or sampling, each available satellite's signal, rather than tracking each continuously.

Solution: A multiplexing receiver is a sequencing receiver that avoids spanning any of the satellite messages' bit boundaries.

Answer is (D)

21. Which of the following are some of the major sources of error in GPS measurement that may be attributed to the GPS receiver?

(A) broadcast ephemeris error
(B) multipath
(C) quartz crystal clock biases
(D) both B and C

Solution: Multipath, the reflection of the satellite signal, and errors of the receiver's on-board oscillator will cause errors in GPS measurements attributable to the receiver.

Answer is (D)

22. What is a cycle slip?

(A) A cycle slip is a reduction in the signal frequency caused by selective availability.
(B) A cycle slip occurs when a satellite's signal is obstructed and cannot be tracked briefly.
(C) A cycle slip is a discontinuity in the P code from a satellite that can be attributed to ionospheric delay.
(D) A cycle slip is a discontinuity in the navigation message from a satellite that can be attributed to tropospheric disruption.

Solution: A cycle slip is caused by a brief obstruction of the signal from a satellite. The receiver is said to "lose lock," and while the measured phase may be the same when the signal from the satellite is reacquired, a number of cycles will have been lost.

Answer is (B)

23. What is the meaning of the word *epoch* as it applies to GPS?

(A) a tabulation of the position of the GPS satellites
(B) the plane of any great circle including a GPS satellite
(C) an interval of time
(D) the cable that connects the GPS antenna to the receiver

Solution: An epoch is an interval of time. The term is most frequently used to describe the incrementation of the moments of data reception.

Answer is (C)

OBSERVABLES

24. What is the minimum number of satellites that must be available for a differential solution of latitude, longitude, height, and time?

(A) 1
(B) 2
(C) 3
(D) 4

Solution: The solution of all four variables requires observation of at least four satellites simultaneously.

Answer is (D)

25. What is a pseudo-range measurement in GPS?

(A) a distance measured between a GPS satellite and receiver based on a time shift that depends on the correlation of the codes each generates to reveal the biased time delay between their respective clocks

(B) a distance measured between a GPS satellite and receiver that depends on the observed phase difference between the carrier signals each generates to reveal the number of wavelengths between the two standards

(C) a measurement of a time series of coordinates using GPS satellites, one stationary receiver, and one roving receiver that depends on the corrected range misclosures between the two receivers

(D) a measurement of the accuracy—that is, the standard deviation—of positions derived from GPS satellites in a particular configuration

Solution: Nearly all civilian receivers that use pseudo-ranges do so with the C/A code. The code is generated in the receiver, derived from its own clock, and is correlated with the same code received from the satellites. The objective of this correlation is to determine the transit time of the signal from the satellites to the receiver. This time multiplied by the speed of light can provide the distance between the satellite and the receiver. The process is complicated by the fact that the clock of the receiver is not synchronous with the satellites' clocks, nor are the satellites' clocks synchronous with one another. These timing errors cause the derived distance to differ somewhat from the actual distance, giving rise to the term *pseudo-range*.

Answer is (A)

26. What is a carrier beat phase measurement in GPS?

(A) a distance between two satellites determined from the comparison of the Doppler-shifted carriers received from both satellites

(B) a measurement of a time series of coordinates using GPS satellites, one stationary receiver, and one roving receiver that depends on the corrected range misclosures between the two receivers

(C) a distance determined by comparing the difference in phase of the carriers generated by two different satellites

(D) a distance between the satellite and receiver determined from the differencing of the reference carrier signal generated in the receiver with the Doppler-shifted carrier signal received from the satellite

Solution: The carrier beat phase is the phase remaining when the received satellite carrier signal is differenced (beat) with the reference frequency generated in the receiver. This is the fundamental tool of the codeless GPS measurement since it often involves squaring the channel to create the second harmonic of the carrier. The codes cannot be retained with this technique, and the method presents a cycle ambiguity that must be solved.

Answer is (D)

27. Which of the following statements describes the elements of a single difference solution in GPS?

(A) Two receivers track the same two satellites simultaneously over several epochs.

(B) Two receivers track two satellites during the same epoch.

(C) Two receivers track the same satellite during the same epoch.

(D) All of the above are true.

Solution: Single difference has the desirable effect of removing clock errors associated with the satellites. Receiver clock errors remain, however.

Answer is (C)

28. Double differencing is the prevailing method of baseline determination. Which of the following statements is true of double differencing?

(A) A double difference is the difference of two single differences.
(B) A double difference involves one satellite tracked at two stations over several epochs.
(C) Two receivers tracking two satellites at the same time is a double difference observation.
(D) Both A and C are true.

Solution: Double differences are a combination of two single differences and can eliminate the effects of the clock errors in both the satellites and the receivers.

Answer is (D)

29. What is the meaning of DOP in GPS?

(A) dilution of precision—a measurement of the geometric strength of the satellite configuration at a particular moment
(B) an abbreviation used to represent the Doppler shift
(C) dynamic operator position—determination of a trajectory with GPS
(D) the standard deviation of the timing errors from the combination of satellite and receiver clocks

Solution: Dilution of precision is a term that predates GPS and refers to a measurement of geometric strength. For example, a GPS position determined when all the observed satellites are bunched together in one part of the sky would not have a good geometric dilution of precision (GDOP). A very good GDOP would be available if one satellite were directly overhead and three others were 15° above the horizon and 120° apart in azimuth at the moment of observation.

Answer is (A)

GPS SURVEYING

30. Considering all the components involved, which of the following is not a factor that affects the accuracy of carrier phase static GPS surveying in the relative mode?

(A) receiver clock bias
(B) modeling of atmospheric refraction
(C) accuracy of satellite orbital data
(D) selective availability

Solution: With station spacings under 100 km, selective availability has no adverse effects. Double differencing techniques can virtually eliminate the satellite clock errors that are the foundation of selective availability. Therefore, selective availability need not affect the accuracy of the relative positions determined by a GPS survey. All of the other aspects mentioned do affect accuracy.

Answer is (D)

31. Generally, what category of precision can be routinely achieved in the measurement of baselines by GPS static relative positioning techniques?

(A) 1 part in 20,000 parts
(B) 1 cm + 2 ppm
(C) 1 part in 100,000 parts
(D) 1 cm + 10 ppm

Solution: The precision that may be attained with any surveying equipment is dependent on the methods used, and GPS is no exception. However, with proper data processing and well-planned observation strategies, precisions of 1 cm + 2 ppm are routinely available. Even this level of precision is somewhat conservative; GPS is capable of 0.3 cm + 0.01 ppm with precise ephemeris information.

Answer is (B)

32. The heights determined by uncorrected GPS measurements refer first to which of the following?

(A) the geoid
(B) a reference ellipsoid
(C) NGVD
(D) mean sea level

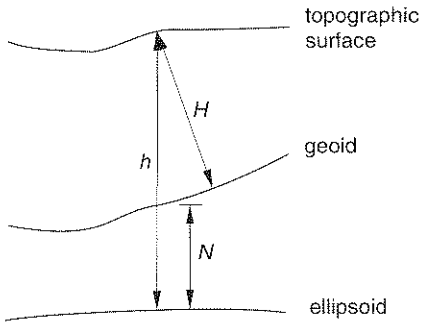
Solution: The heights determined by GPS measurements are originally ellipsoidal. These heights are then generally converted to orthometric heights, also known as *elevations*, by modeling the differences between the geoid and reference ellipsoid at the newly observed stations, using connections to control points with known orthometric heights.

Answer is (B)

33. Which of the following formulas most closely describes the relationship between ellipsoidal height, h, geoidal height, H, and geoidal undulation, N?

(A) $h = H - N$
(B) $H = \dfrac{h}{N}$
(C) $N = h + H$
(D) $h = H + N$

Solution: The ellipsoidal height is the sum of the geoidal height and the separation between the ellipsoid and the geoid, also called the *geoidal undulation.* The calculation does presume there is no significant deviation of the vertical. In the coterminus U.S., the geoid is actually below the ellipsoid, giving N a negative value.



$$h = H + N$$

Answer is (D)